A method of measuring mediolateral patella maltracking through the use of stereoscopic 3D imaging cameras.

Introduction:

Patellofemoral pain is experienced by ~23% of the general population during their lifetime (1), and it is suggested that if left untreated can cause cartilage degeneration, leading to osteoarthritis and poor joint function (2,3). A leading cause of patellofemoral pain is patella maltracking (4), and while orthopaedics frequently diagnose patella maltracking, they do not quantitatively measure its extent.

This project addresses that gap by developing a design specification for a knee-mounted device capable of measuring patellar motion in vivo through 3D stereo vision. This project investigated the use of an Intel® RealSense™ D405 camera to measure the mediolateral position of the patella relative to the transepicondylar axis of the femur between 0-90° flexion. Since patella tracking results from the complex interaction of anatomical geometry and muscular activity (4), accurate diagnosis must take place under dynamic conditions and various load cases. The patella's motion will be measured *in vivo* during weight bearing and non-weight bearing movements.

The design adheres to relevant ISO standards to ensure accuracy and repeatability, outputting patellar displacement against knee flexion angle to support orthopaedic assessment of maltracking severity.

Biomechanics overview:

The patella is a sesamoid bone embedded within the quadriceps tendon (5). It exhibits 6 degrees of freedom in motion, of these 4 can be deemed to be clinically significant motion variables: flexion tilt, rotation, and shift (6), as seen in ***figure 1***. Among the patellar motion criteria, mediolateral translation (shift) is often central descriptor of maltracking. Imaging studies often define patellofemoral maltracking primarily as abnormal mediolateral translation of the patella relative to the trochlear groove, and dynamic MRI studies shows that this translation is markedly increased in symptomatic patients compared with controls (7,4). Patella maltracking can be classified into distinct grades based on the magnitude of subluxation: mild <5mm; moderate 5-10mm and severe >10mm (7). This motion is often defined relative to the anatomical axis of the femur but can also be defined relative to the direction parallel to the transepicondylar axis (TEA), an axis between a line between the most medial and most lateral prominences of the epicondyle (8) as shown in ***figure 1***

The knee joint is not a typical hinged joint but a four-bar linkage (9). The patella acts as a fulcrum, to increase the mechanical advantage of the quadriceps muscles. The result is a large amount of force flows through the patella (10). During normal tracking, the patella is centrally aligned within the trochlear groove, and tracks with the joint between 20-90°of flexion (11). This distributes force over the full surface area of the patella.

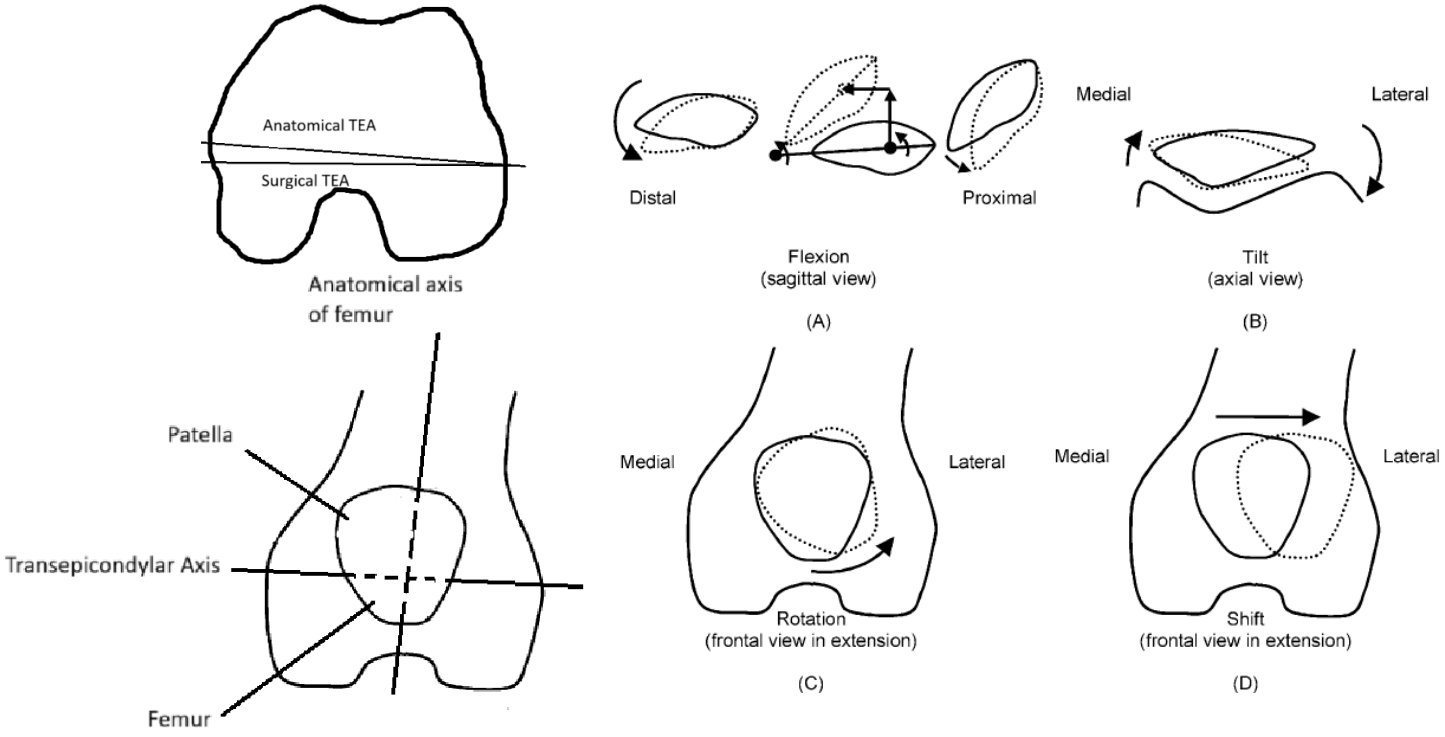


Figure 1: diagrams detailing types of patella motion, and relevant anatomical axis (6)

Patella maltracking is problematic for the proper function of the knee, particularly during load-bearing activities, such as squatting and climbing stairs. At flexion angles between 0° and 90° patellofemoral joint reaction forces and cartilage stresses are high and symptoms are typically aggravated (12). This is because patella shift decreases the surface area of the patella in contact with the articular cartilage of the trochlear groove, causing an increase in stress which can lead to anterior knee pain and cartilage degradation; a serious and damaging condition that can develop into osteoarthritis (2). Therefore, early intervention is key, this is often in the form of patella tracking knee braces like the Ossur Formfit Tracker® in conjunction with stretching and strengthening exercises that target the muscles responsible for the imbalance (13). To support early diagnosis and ongoing monitoring, this project proposes a suitable design specification for a device that can detect mediolateral shift relative to the TEA.

3D stereo camera working principles:

To accurately observe the position of the patella, image processing of the data obtained stereo cameras offer a valuable solution.

Stereo cameras work much like human binocular vision to perceive depth. The parallax difference due to the separation of the two camera sensors can be used to calculate the disparity between the images of either camera, this is then used to calculate depth. Stereo image processing follows a series of mathematical

procedures to convert a pair of 2D images into a 3D point cloud. The process is typically carried out in this order: **calibration, rectification, correspondence, triangulation** and finally **3D reconstruction,** a process that follows a project light ray path, to locate a point in space (14).

Some key limitations of stereo imaging are texture and distance. If pixel properties are too uniform, the image cannot be correctly corresponded and depth data will not be outputted accurately. For the design's application, skin on the knee has a lot of texture and colour variation, making this not a major concern. Stereo camera precision is inversely proportional to depth meaning at large distances small disparity errors lead to large depth errors (14).

Component Selection:

The design specification table, found at the bottom of this document, defines four main subsystems: a mechanical rig to maintain a fixed relationship between the camera and the femoral transepicondylar axis, a sensor to measure knee flexion angle between 0–90° with ±2° accuracy, a sensor to track patellar position in three dimensions, and an electronics unit to acquire and transmit data at 60 Hz. The overall system must achieve an accuracy <3mm to be clinically feasible in quantifying moderate and severe maltracking

A functional knee brace can be used to constrain the camera at a known position relative to the TEA. This is favourable to mounting the camera on an external tripod, as the brace will keep the camera fixed distance ~ from the knee during dynamic movement, the rig will also ensure the patella remains within the cameras Field of View (FOV) at flexion angles between 0-90°. Functional knee braces are not restrictive to anterior knee motion and therefore do not alter patellar kinematics. This brace can be 3D printed using PLA, which when tested by BCN 3D technologies following ISO 527 and ISO 1133 its mechanical properties were found to be: E= 3.12GPa; σ_y= 70MPa (15). The stress through the brace must not exceed the yield stress. By assuming an extreme use case where the hinge supports the full bodyweight of a 90kg user, the required material cross-sectional areal can be determined with a safety factor of 2, the minimum required cross-sectional area at this hinge is required to be.

$$A_{min} = \frac{2mg}{\sigma_y} \approx 25.2mm^2$$

The calculated value lies well below the available section area in the design, confirming sufficient structural strength. SolidWorks volume analysis at 100% infill gave a total mass of 300 g.

Patella tracking sensor: 3D cameras versus IMUs:

Inertial measurement units (IMUs) where considered as a lower cost alternative, however, were deemed to not produce results to the required accuracy. IMUs measure orientation and linear acceleration, requiring double integration to estimate their position, and must be attached directly to the skin. Their readings are therefore affected by soft-tissue artefact, which varies substantially between individuals and can exceed several millimetres, surpassing the ±3 mm accuracy threshold.

A short-range stereo camera mounted on the brace avoids these issues by directly reconstructing the anterior knee surface as a 3D point cloud. The Intel® RealSense™ D405 provides a depth accuracy < 1 mm at 70 mm distance and a working range of approximately 70–500 mm, with a frame rate of 60 Hz (16). Operating the camera at a fixed distance of 105mm, keeps the patella within the region of highest depth precision while maintain an FOV sufficient to capture motion over the full flexion range. The point cloud can be processed to extract the mediolateral patella position in mm relative to its initial position at 90 degrees flexion, with is motion calculated relative to the TEA.

Knee flexion angle measurements and electronics:

Knee flexion angle is measured by a 10 kΩ rotary potentiometer integrated into the lateral hinge. The sensor operates at 60 Hz, matching the camera frame rate, and achieves angular accuracy of ±2° across the 0–90° range. It is connected to an ESP32 microcontroller equipped with a 12-bit analogue-to-digital converter and powered by a 3.7V 2000 mAh lithium-ion battery through a battery management system. The microcontroller supplies power to the potentiometer and streams synchronised data to the host computer via USB-C alongside the stereo camera. Acquisition software aligns timestamps between the two inputs to produce a continuous time series of patellar displacement versus flexion angle. The combined mass of the electronic components is approximately 184g, giving a total device weight of 484g, below the 500g limit specified in the design requirements.

Measuring Patella shift using the device:

The clinician first identifies the transepicondylar axis (TEA) by palpating the medial and lateral femoral epicondyles and marking the lateral and medial extremes with a skin-safe marker. The hinge axis of the brace is then aligned with the TEA, and the adjustable Velcro straps are tightened to secure the device. To aid visual alignment, the clinician locates the most lateral point of the patella by palpation and marks it with a contrasting dot.

The system is calibrated in two static postures: full extension (0°) and deep flexion (90°). These positions define the reference transformation between the knee and camera coordinate frames. The subject then performs five flexion–extension cycles between 0° and 90° to establish repeatability, followed by weight-bearing trials such as walking or squatting to assess dynamic tracking performance.

The software processes each frame sequence to output a graph of knee flexion angle versus patellar position. Patellar displacement is defined relative to the lateral reference point at 90° flexion, with positive values indicating lateral motion and negative values indicating medial motion along the TEA. Because all coordinates are expressed relative to the brace's initial calibration, small translational movements of the rig do not affect results provided the patella remains within the camera's field of view throughout the trial.

Design risks and limitations:

The motion of the human knee exhibits a moving instantaneous centre of rotation rather than a fixed hinge axis. Despite the hinge of the brace being initially aligned with the TEA the difference in rotation centres

will result in relative motion between the brace and the TEA. As a result, the assumed camera position relative to the TEA will not be truly fixed and may translate several millimetres. This misalignment introduces a systematic error to any analysis that assumes a constant reference frame, and if severe enough, the patella could move out of the stereo cameras FOV and minimum recording distance, leading to data gaps and inaccurate results. Although a polycentric hinge could help mitigate this, the lack of standardisation in moment arm length and patella size will still cause issues.

The added mass of the device (~0.48 kg) on the lateral aspect of the leg may subtly alter gait or induce a small valgus moment, although this effect is expected to be minimal for static and slow dynamic trials

Another limitation is due to the camera not imaging the patella directly but the overlying skin and subcutaneous tissue. Soft tissue artefact, defined as relative motion between skin and underlying bone, is a well-recognised source of error in optical measurements of knee kinematics (17,18). MRI and radiographic studies report prepatellar subcutaneous fat thickness' ranging from around 5–6 mm in non-obese individuals to ≥15 mm in obese and morbidly obese cohorts (19,20). Thus, in obese patients the patellar bone surface may lie more than 1–2 cm beneath the skin, smoothing the external contour and markedly increasing the separation between bone and the measured surface.

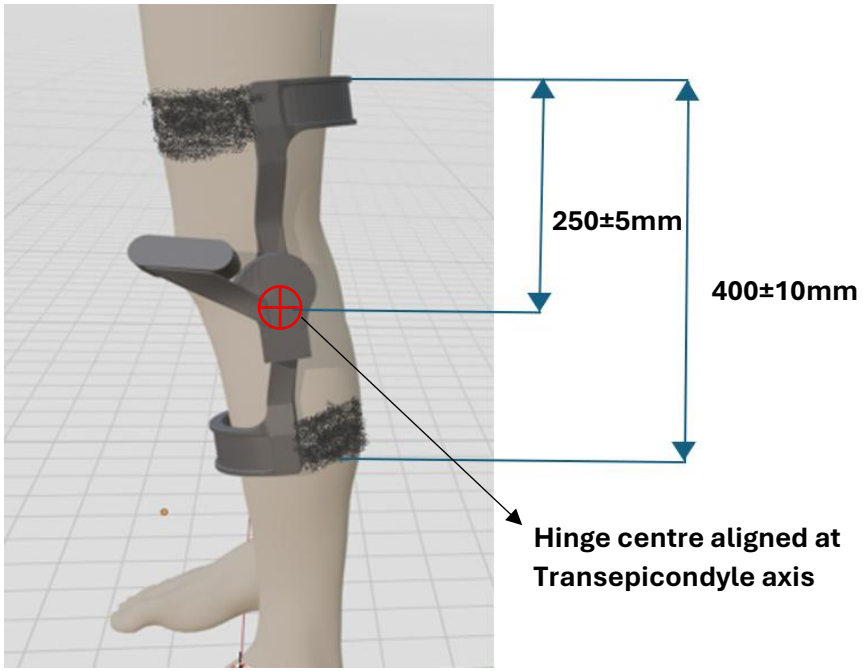
This implies that surface-based 3D tracking may recover skin motion rather than true patellar kinematics in these individuals, regardless of the sophistication of the tracking algorithm, decreasing its accuracy. As such, the device is unable to provide precise patella tracking for obese patients. This is a key limitation as patients with patella femoral pain and patella maltracking are likely to be overweight or obese, with 38% of young adult patients with this condition are overweight or obese (21). On top of this, patients with a thigh diameter greater than 175mm or a calf diameter greater than 150mm or patients showing valgus or varus angles greater than 8 degrees will not be able to use the brace due to geometric constraints.

Finally, the absence of a high-fidelity reference such as dynamic MRI or fluoroscopy limits validation of the system's absolute accuracy. The device therefore provides relative tracking performance rather than a definitive measurement of bone motion. Future iterations should incorporate polycentric hinges or calibration markers to reduce reference drift and improve the robustness of dynamic patellar tracking.

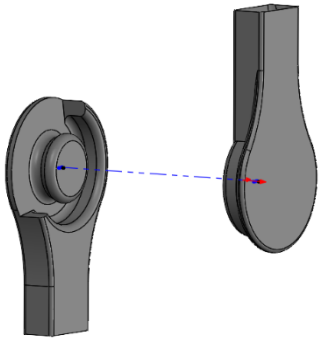
Conclusion:

This project established the design specification, user procedure, and feasibility of a knee-mounted stereo-camera system for tracking patellar motion between 0–90° flexion, under both load-bearing and non-load-bearing conditions using ISO standards and scientific literature. The design specification fixes an Intel® RealSense™ D405 stereo camera to a functional knee brace, maintaining a fixed 10.5 cm camera-to-knee distance, with an alignment tolerance of ±2mm to the transepicondylar axis, and tracking accuracy within <3 mm, exceeding the 5mm requirement. The brace, 3D printed using PLA can withstand expected load cases with safety factor of 2. It provides adjustable fit for thigh and calf diameters of 145–175mm and 120–150mm. Key limitations include hinge misalignment, optical occlusion, and patient BMI.

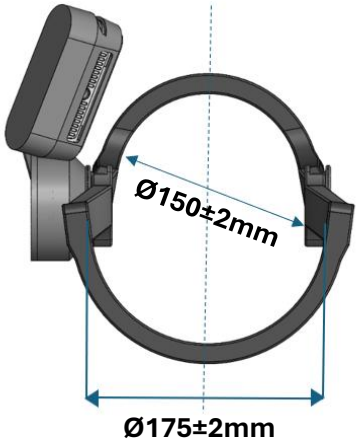
Side View (a)



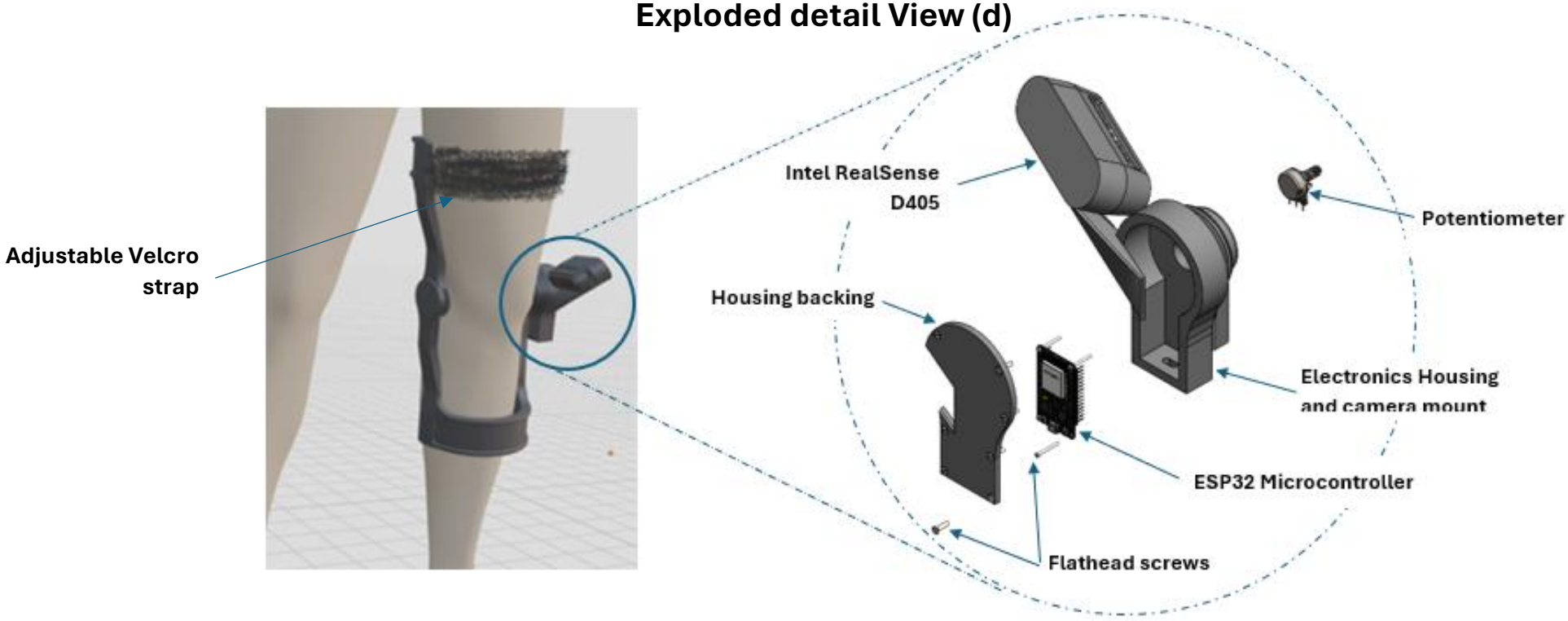
Hinge exploded view (b)



Top View (c)



Exploded detail View (d)



Design specification table

User	Need	Implication	References	Calculation/ test	Design requirement
Practitioner	Measure patella motion	The position of the patella at a given time needs to be quantitatively measured, patella maltracking needs to be accurately described	A systematised MRI approach to evaluating the patellofemoral joint. (7)	Compare measured patella trajectory with reference MRI or motion capture data	Accuracy of patella tracking must be within ±3 mm over the range of motion
Device	Stereo camera must capture the motion of the patella from 0-90° flexion	The knee must be within the cameras field of view, and minimum tracking depth for entire ROM	ISO 20685-1: 2018 3d scanning. Intel RealSense data sheet	Minimum separation = minimum recording distance + buffer = 7cm+50%= 10.5cm; Trigonometry using FOV angle of 120	The camera rig must hold the camera a minimum of 10.5cm away from the knee
User	comfortable fit around the legs	The diameter of the thigh and calf section of the brace should be an appropriate size for a range of users	BS EN ISO 7250 (Basic human body measurements)	Measure thigh and calf circumference across 5th–95th percentile user	Adjustable range of ±30 mm for both thigh and calf circumference. Using Velcro straps
User	no translation relative to the user's legs during motion	The device must remain fixed throughout movement	--	Dynamic fitting test with 0–90° flexion; check for displacement	Use adjustable Velcro straps, to hold brace in place.
User	The device should not damage the user's skin	The material used should not be abrasive	BS EN ISO 10993-10 (Skin irritation); Material datasheets	Abrasion and comfort test using soft tissue phantom, looking for swelling or grazing	Surface roughness Ra ≤ 0.8 μm; hypoallergenic material required
Practitioner	Record knee flexion angle data	Must record angular position of joint in real time	Potentiometer spec sheet	Calibrate potentiometer using 0°, 45°, 90° physical reference points	contain a potentiometer that is calibrated and a microcontroller to read its output
Practitioner	Intuitive data should be outputted	Output should be user-friendly and interpretable	ISO 9241-210:2019 (Human-centred design)	Receive feedback from practitioners	Real-time digital display (latency <0.5 s) with angle and patella coordinates
Device	The hinge of the brace should be aligned with the transepicondyle axis	The transepicondyle axis needs to be determined marked to check alignment	Anatomical alignment literature (Grood & Suntay, 1983)	Visual marking and alignment jig test	Hinge must align with transepicondyle axis within ±2mm
Device	Must not break due to forces during testing	Material must be able to withstand peak stress.	BS EN ISO 10328 (Structural testing of orthoses)	Finite element simulation of 500 N load; physical stress test	Yield strength ≥ 2× expected operating stress
Practitioner	Measurements should be between 0-90° of flexion	The device should not allow hyperextension (flexion angle below 0°) and the hinge should move at least 90°	--	Angular limit test (mechanical stops)	Mechanical stops must prevent extension below 0°
User	The device should not restrict motion	The device should be comfortable, allow the user to move freely, it should take into around the biomechanics of the user	ISO 6385:2016 (Ergonomics principles in the design of work systems)	Test client walking with and without brace and ensure the walk is symmetrical	The device should not weigh more than 0.5kg, the device should not get in the way of the user when performing dynamic activities. A battery should be used so cables do not obstruct the user.

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